

Material Safety data sheet

1. Substance/preparation and company identification

E-129

- Use: Detergent and cleaning agent for the chemical industry
- Company: Bifrost Chemtech Co.,Ltd
Telephone: 02-117-1984
Telefax number: 02-117-1986

2. Composition/information on ingredients

Chemical nature : Fatty acid ethoxylate

3. Hazard identification

Irritating to skin.

4. First-aid measures

- General advice: Immediately remove contaminated clothing.
- On skin contact: Wash thoroughly with soap and water.
- On contact with eyes: Wash affected eyes for at least 15 minutes under running water with eyelids held open.
- On ingestion: Rinse mouth immediately and then drink plenty of water, seek medical attention.

5. Fire-fighting measures

- Suitable extinguishing media: water, dry extinguishing media, foam, carbon dioxide
- Specific hazards: harmful vapours. Evolution of fumes/fog. The substances/groups of substances mentioned can be released in case of fire.
- Special protective equipment: Wear a self-contained breathing apparatus.
- Further information: The degree of risk is governed by the burning substance and the fire conditions. Contaminated extinguishing water must be disposed of in accordance with official regulations.

6. Accidental release measures

- Personal precautions: Use personal protective clothing.
- Environmental precautions: Contain contaminated water/firefighting water. Do not discharge into drains/surface waters/groundwater.
- Methods for cleaning up or taking up: For large amounts: Pump off product. For residues: Pick up with suitable absorbent material. Dispose of absorbed material in accordance with regulations.

- Additional information: High risk of slipping due to leakage/spillage of product.

7. Handling and storage

Handling

- No special measures necessary provided product is used correctly.
- Protection against fire and explosion:
- Take precautionary measures against static discharges.

Storage

- Further information on storage conditions: Keep container tightly closed and dry; store in a cool place.

8. Exposure controls and personal protection

Personal protective equipment

- Respiratory protection: Respiratory protection in case of vapour/aerosol release.
- Hand protection: Chemical resistant protective gloves (EN 374)
- Eye protection: Safety glasses with side-shields (frame goggles) (EN 166)
- Body protection: Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to DIN-EN 465).
- General safety and hygiene measures: Handle in accordance with good industrial hygiene and safety practice. Wearing of closed work clothing is required additionally to the stated personal protection equipment.

9. Physical and chemical properties

Form :	liquid
Colour :	yellowish Clear
Odour:	product specific
pH value :	approx. 6-8 (5% , 25°C)
Flash point :	> 100 °C
Density :	approx.1.02 g/cm ³ (23°C)
Solubility in water :	dispersible

10. Stability and reactivity

- Hazardous reactions:
No hazardous reactions when stored and handled according to instructions.
- Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

11. Toxicological information

- Actual
Oral : No information available
Skin irritation : No information available
Eyes irritation : No information available
In vitro mutagenicity : No information available

12. Ecological information

- Environmental fate and transport
Biodegradation : No information available
- Environmental toxicity
Acute toxicity to aquatic invertebrates : No information available
Toxicity to aquatic plants : No information available
Other ecotoxicological advice : No information available

13. Disposal considerations

Must be dumped or incinerated in accordance with local regulations.

Contaminated packaging:

Uncontaminated packaging can be re-used.

Packs that cannot be cleaned should be disposed of in the same manner as the contents.

14. Transport information

Not classified as hazardous under transport regulations

15. Regulatory information

- Hazard symbol(s)
Xi : Irritant.
- R-phrases
R38 : Irritating to skin.
R50 : Very toxic to aquatic organisms.
- S-phrases
S26 : In case of contact with eyes, rinse immediately with plenty of water and seek medical advice.
S39 : Wear eye/face protection.

16. Other information

Information on intended use: This product is of industrial quality and unless otherwise specified or agreed intended exclusively for industrial use. Any other intended applications should be discussed with the manufacturer. This includes the mentioned and recommended usage. In particular this concerns the application for products that are the object of special standards and regulations.